

7 Core Tabs
ADMIN CONTROL CENTER

< 0.05s Sync
REAL-TIME CLOUD PUSH

BCrypt Salted
IRREVERSIBLE SECURITY

1st of Month
AUTOMATED ACCRUAL

1. Admin Profile, Security & Account Settings (Tab 7)

7 Key Capabilities

01

Director Identity & Public Channel Setup: Configure official educator full name, designation/role, official mobile, and email address.
How to use: Open Admin Settings tab → Edit Director Name/Email/Mobile → Authorize with current password.

PROFILE

02

Director Avatar Photo Upload & Sync: 64px circular gradient preview with 1-click camera upload, auto-compression, and real-time cloud sync.
How to use: Click "Change Photo" button → Select JPG/PNG → Image is auto-saved and synced.

AVATAR

03

Institutional UPI ID (VPA) Configuration: Sets default receiving UPI ID (e.g. chandankr1501998@ybl) printed across all fee receipts and payment QR codes.
How to use: Enter UPI VPA in Card 1 → Saves to cloud and automatically updates all student payment vouchers.

UPI VPA

04

Irreversible Salted Password Hashing: Admin and student passwords are encrypted into 60-char BCrypt hashes (\$2a\$10\$...) that no one can view or decrypt.
How to use: Type new password (min 12 chars) → BCrypt generates secure salt & hash automatically.

BCRYPT

05

Inline Password Visibility Toggles: Dedicated eye-toggle buttons securely anchored inside input fields to easily verify complex passwords before saving.
How to use: Tap the eye icon on the right side of password inputs to reveal or hide password text.

EYE TOGGLE

06

Mandatory Security Authorization Step: Red security verification box requiring current password entry to prevent unauthorized profile/credential hijacking.
How to use: Enter current password in the red verification box before clicking "Save & Sync Changes".

MANDATORY

07

Permanent Read-Only Admin Identifier: Permanent admin username (e.g. chandan) locked in read-only mode to prevent credential corruption or lockout.
How to use: Permanent institutional username is preserved safely across all browser sessions and cloud syncs.

FIXED ID

2. Student Roster, Admissions & Fee Recording (Tab 1)

8 Key Capabilities

08

Sequential Collision-Proof Student ID Generator: Generates unique IDs (e.g. 261001: Year 26 + Class 10 + Seq 01) with zero collisions under concurrent admissions.
How to use: Click "Add New Student" → ID is generated automatically based on selected class batch.

YYCCSS

09

Segmented 4-Preset Fee Recording Modal: 4-in-1 collection modal with 2x2 quick amount presets (₹500, ₹1,000, ₹2,000, Full Due) and Cash/UPI channel tagging.
How to use: Click "Pay" on student row → Tap preset or type custom amount → Select Educator & Cash/UPI → Submit.

PAYMENT

10

Instant Printable Fee Receipt Generation: Auto-creates official printable PDF receipts with unique numbers (REC-XXXX), student metadata, and director signatures.
How to use: Click "Print Receipt" from student row or modal to print or download PDF voucher instantly.

PDF VOUCHER

11

Post-Admission Rate & Concession Adjustments: Adjusts monthly tuition rates, applies merit fee concessions, or invoices add-on test series without corrupting cash totals.
How to use: Click "Adjust Fee" → Choose "Rate Revision" or "Fee Concession" → Enter adjustment note → Save.

CONCESSION

12

1-Click DOB Password Reset for Students: Clears student custom password hash and resets login access back to default DOB (YYYY-MM-DD) in 1 second.
How to use: Click "Reset Password" button on any student row → Confirms and resets login to DOB.

RESET

13

Financial Deletion Guardrails & Cascade Purge: Strict safety protection blocking student deletion if unpaid dues exist. Cascade-cleans ledger upon verified purge.
How to use: Dues must be ₹0 to delete student, ensuring complete financial accountability.

GUARD

14

Multi-Field Search & Dynamic Filter Bar: Instant real-time search filtering across Student Name, Roll Number, Guardian Mobile, and Due Status (All / Dues / Cleared).
How to use: Type in search box or click status pills ("Has Dues" / "Cleared") for instant roster filtering.

SEARCH

15

Batch-Wise Partitioning (10th / 9th / 8th / Junior): Horizontal segmented pills isolating Class 10th Achievers, Class 9th Nurture, Class 8th Alpha, and Junior students.
How to use: Tap batch pills above the table to view isolated batch rosters and batch dues totals.

BATCHES

3. Revenue Analytics & Educator Collection Breakdown (Tab 2)

5 Key Capabilities

16

Chandan Kumar Revenue Collection Attribution: Detailed accounting separating total cash and UPI collections received directly by Science Lead Chandan Kumar.
How to use: Open Analytics tab → View "Chandan Kumar Collections" card showing Cash + UPI + Grand Total.

CHANDAN SIR

17

Prof. Ravi Ranjan Revenue Collection Attribution: Detailed accounting separating total cash and UPI collections received directly by Mathematics Lead Prof. Ravi Ranjan.
How to use: Open Analytics tab → View "Prof. Ravi Ranjan Collections" card showing Cash + UPI + Grand Total.

RAVI SIR

18

100% Reconciled Financial Synthesis: Guarantees that the sum of Teacher Collections (Chandan Sir + Ravi Sir) matches 100% with Total Institute Revenue.
How to use: Checked automatically in real time across master ledger entries and admission fee records.

100% MATCH

19

High-Due Defaulter Prioritization Engine: Identifies students with pending dues exceeding 1 month and organizes them in descending order of outstanding balance.
How to use: Click "View Defaulters" in Analytics to see prioritized list with 1-click WhatsApp/Call buttons.

DEFAULTERS

20

Class-Wise Fee Collection Radial Visualizer: Visual percentage distribution comparing fee collections across Class 10th, 9th, 8th, and Junior Foundation batches.
How to use: Visual gauges update live whenever a payment or concession is recorded in the portal.

CHARTS

4. 1st-of-Month Unified Billing & Auto-Reminder Schedule

5 Key Capabilities

- 21



1st-of-Month Unified Fee Accrual (Day 1 to 4): Automated scheduler accrues monthly fees (₹800 for 8th, ₹1,000 for 9th/10th) across all active batches on the 1st of every month.

DAY 1-4

 **How it works:** Runs automatically on the 1st of each month or can be triggered manually by clicking "Run Monthly Billing".
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Billing Idempotency & Duplicate-Run Shield: Locks each month with a unique hash (accrued_YYYY-MM) so triggering billing multiple times never double-charges students.

IDEMPOTENT

 **How it works:** If billing already ran for "2026-08", repeat runs are safely ignored without balance corruption.
- 23



Mid-Month Due Reminder Cycle (Day 15 to 19): Automated reminder engine identifying students who still have unpaid dues and queuing polite payment alerts for parents.

DAY 15-19

 **How it works:** Evaluates pending balances on Day 15 and prepares formatted WhatsApp/Email payment vouchers.
- 24



1-Click Manual Billing Trigger Button: Admin button allowing directors to instantly apply monthly fee accruals on-demand for all batches with live progress toast.

MANUAL RUN

 **How to use:** Open Admin Settings or Analytics → Click "Trigger Monthly Fee Accrual" → Confirms and updates dues.
- 25



Automated Billing Ledger Entry Generation: Creates itemized ledger entries (MONTHLY-YYYY-MM) documenting billing timestamp, batch rate applied, and updated balance.

HISTORY

 **How to use:** Visible in each student's fee history voucher and in the master institutional audit trail.

5. Resend Live Email Engine & Notification Center (Tab 3)

5 Key Capabilities

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Resend Live Transactional Email Dispatcher: Connected directly to Resend API sending official HTML emails from verified domain @pragyaninstitute.com.

RESEND API

 **How to use:** Open "Email Dispatcher" tab → Enter recipient or select batch → Type subject/body → Click Send.
- 27



Instant Payment Receipt Emailing: Automatically emails styled fee receipts to student/parent email addresses upon recording any cash or UPI payment.

RECEIPTS

 **How to use:** In the Payment modal, check "Email Receipt to Student" → Dispatches receipt HTML instantly.
- 28



Batch-Wide Announcement Email Broadcasts: Send test schedules, holiday notifications, and exam alerts to entire class batches simultaneously in 1 click.

BROADCAST

 **How to use:** In Email Tab, choose "Target: Class 10th" or "All Students" → Enter message → Dispatches in bulk.
- 29



Dynamic Personalization Template Tags: Merges student dynamic tags (e.g. {{student_name}}, {{roll_no}}, {{due_amount}}, {{upi_link}}) into email body.

TEMPLATES

 **How to use:** Use preset template buttons ("Fee Due Reminder", "Exam Alert") for auto-filled HTML emails.
- 30



Invalid Email Auto-Filter & Safety Guard: Validates recipient email formatting before sending to prevent API bounce penalties and maintain high sender reputation.

BOUNCE-SAFE

 **How to use:** Automatically validates email strings and reports real-time delivery status in UI toasts.

6. Student Requests, Campus Notices & Audit History (Tabs 4, 5, 6)

5 Key Capabilities

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Student Profile Request Approval Queue (Tab 4): Review submitted student profile changes with side-by-side comparison of old vs. new values before approving.

REQUESTS

 **How to use:** Open Requests Tab → Click "Approve & Update" to accept or "Reject" to decline with note.
- 32



Notice Publishing & Category Management (Tab 5): Create campus bulletins with tags (Exam, Holiday, Urgency), target specific batches, and pin urgent alerts.

NOTICES

 **How to use:** Open Post Notice Tab → Enter Title, Category, and Details → Click "Broadcast Announcement".
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Searchable Institutional Audit Trail (Tab 6): Immutable history logging every educator action (Admissions, Payments, Resets, Deletions) with timestamps.

AUDIT LOG

 **How to use:** Open Audit History Tab → Filter by Educator (Chandan Sir / Ravi Sir) or search student roll.
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Floating Real-Time Action Feedback Toasts: Universal top-layer feedback engine displaying green success, yellow warning, or red error alerts on every action.

NOTIFICATIONS

 **How to use:** Automatically gives immediate visual confirmation upon saving, deleting, or recording data.
- 35



Audit Log Retention & Security Archival: Audit events are permanently preserved to prevent tampering, ensuring complete transparency for institute directors.

PURGE SAFETY

 **How to use:** Directors can review complete historical timelines of all fee modifications anytime.

7. Sub-50ms Cloud Sync, Offline Resilience & Infrastructure

5 Key Capabilities

- 36



Sub-50ms Supabase Postgres Cloud Sync: Real-time bi-directional database synchronization keeping offline browser state and cloud PostgreSQL updated within 0.05 seconds.

< 0.05s

 **How it works:** All changes made on admin phone or laptop sync to cloud PostgreSQL instantly.
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Offline Mutation Queue & Network Auto-Recovery: Stores offline edits locally during internet drops and automatically flushes and reconciles changes upon reconnection.

OFFLINE

 **How it works:** You can record payments or take attendance even without internet; syncs when back online.
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Google Search Console & Live Sitemap Integration: Verified domain property with live XML sitemap (/sitemap.xml) and robots.txt crawler directives.

GOOGLE

 **How it works:** Automatically keeps pragyaninstitute.com indexed and visible in Google Search.
- 39



Google Knowledge Graph Educational Schema: Rich EducationalOrganization and LocalBusiness JSON-LD markup driving #1 search rankings on Google.

SCHEMA

 **How it works:** Informs Google Knowledge Graph of faculty, address, and official domain for top rank.
- 40



41/41 Master Automated CI/CD Test Suite: Complete test suite covering authentication, billing idempotency, concurrency, and security with 100% pass rate.

TESTS

 **How it works:** Validated automatically before every build to guarantee zero runtime bugs.